Nallabati Venkata Durga Sai Sarat Chandra

Target Role: Robotics & Controls Engineer Intern | Ather Energy

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SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive software architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech), *Amrita Vishwa Vidyapeetham* — **2024 – Present**
CGPA: 9.05 / 10.00 — *Amritapuri*

Higher Secondary Education, *Sri Chaitanya Vidhyanikethan* — **2024**
CBSE (MPCEA) Percentage: 88.2% — *Vishakapatnam*

Secondary School Education, *Sri Chaitanya School* — **2022**
SSC AP Marks: 570 / 600 — *Rajam*

SKILLS

Programming Languages: C++ (C++17/C++20), Python (3.12), C (C99/C11), Bash, SQL, POSIX Sockets, Multi-Threading

Software Architecture & Design: Object-Oriented Design (OOP), Data Structures & Algorithms, REST APIs, Microservices, Asynchronous UDP/TCP Sockets

Robotics & AI Infrastructure: ROS 2 (Jazzy), CycloneDDS, OpenCV (35+ FPS), PyTorch, ONNX Runtime, Vitis AI EP, Zero-Copy Shared Memory

Tooling & CI/CD Systems: Linux (Ubuntu Server), Git/GitHub, Docker, CI/CD Pipelines, PyTest, GDB, Valgrind, CMake, Tectonic LaTeX

PROJECTS

Pasupatha Autonomous Companion Robot Control Bus — **Head of Embedded Systems, Club Mirai**
ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed
- 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

Autonomous Mobile Robot (AMR) Edge Platform (IEEE ICRM 2025) — **First Author & Lead Developer**
ARM Cortex-A, OpenCV, Probabilistic Hough Transforms, Asynchronous UDP Sockets
- 35+ FPS lane boundary tracking; <15ms perception-to-actuation telemetry latency.

Evolution Edge: Self-Evolving Heterogeneous Neural Bridge — **Team Lead, AMD Developer Hackathon**
AMD Ryzen AI NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, INT8 PTQ, PEFT/LoRA
- Sub-28.4ms deterministic inference @ 8W TDP; offloaded tensor compute across integrated GPUs/NPUs.

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity, *IEEE ICRM 2025* — **11/2025**
- Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.

ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.